

Technical Assessment Report: Time Series Forecasting & Computer Vision

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TO: Dr. Sohn's Research Lab, PhD Technical Assessment Review
FROM: Sampada Kharel (sampadha.kharel@gmail.com)
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RE: Part 1: Capital Bikeshare Demand Analysis & Forecasting; Part 2: Building Footprint Segmentation (full code: *part1_bikeshare.ipynb*, *part2_segmentation.ipynb*)

Part 1: Capital Bikeshare Demand Analysis & Forecasting

Data. UCI Bike Sharing dataset (17,379 hourly records, Washington D.C., 2011–2012). Weather variables were denormalized to physical units; season codes were corrected to match the observed calendar months, since the raw coding is off by one step from the actual dates.

Descriptive statistics & exploratory analysis

- Demand is overdispersed (mean ≈ 189 , variance $\approx 32,900$ for hourly count), ruling out a simple Gaussian or equidispersed-Poisson model.
- Workdays show a sharp bimodal commute pattern (peaks at 08:00 and 17–18:00); weekends/holidays show one broad midday peak.
- Casual (leisure) demand is far more weekend- and weather-sensitive than registered (commuter) demand; casual share of total rentals swings from $\approx 9\%$ at commute hours to $\approx 26\%$ in the afternoon.
- Demand grew 63% from 2011 to 2012, requiring a trend/year term in any forecasting model.

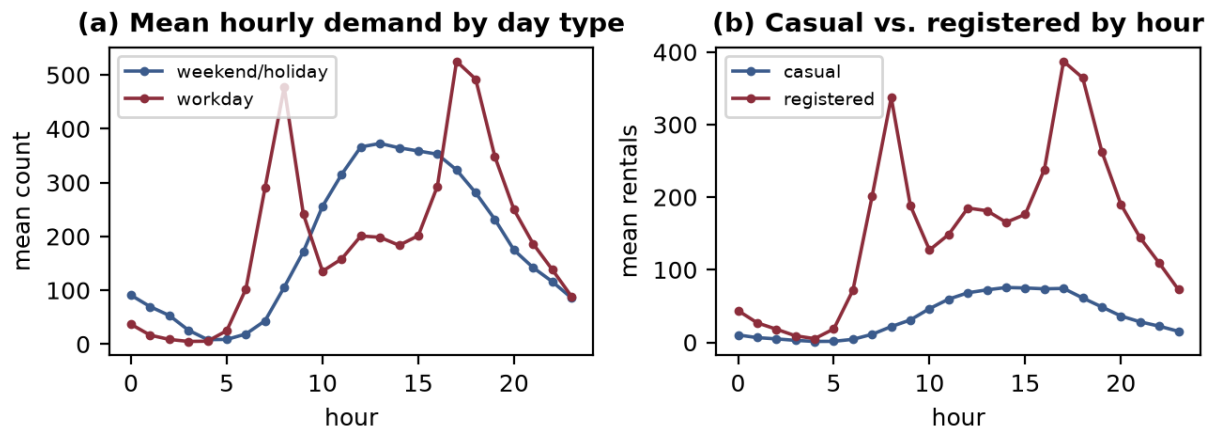


Figure 1. Hourly demand by day type, and casual vs. registered by hour.

Statistical & regression analysis

Models were built in stages to address each statistical issue directly: an OLS model on $\log(\text{count}+1)$ surfaces multicollinearity and non-linearity; moving to a Negative Binomial GLM addresses the count-data distribution.

Diagnostic	Result	Action taken
Multicollinearity (temp vs. atemp)	VIF 43.6 / 43.7 together, 1.0 apart	Dropped atemp
Non-linearity (temperature)	AIC improves 31,260 to 30,846 adding temp ²	Quadratic term retained
Hour x workingday interaction	AIC improves to 14,755 adding it	Interaction retained (F-test $p \approx 0$)
Count distribution	Overdispersion confirmed (Cameron-Trivedi test, $p \approx 0$)	Negative Binomial GLM used
Residual autocorrelation	Durbin-Watson = 0.83	HAC (Newey-West) standard errors used

Casual and registered users were modeled separately as the assessment allows, using incidence rate ratios (IRR) from a Negative Binomial fit.

Term	Casual IRR	p	Registered IRR	p
workingday	0.45	<0.001	0.99	0.092
holiday	0.82	<0.001	0.79	<0.001
temp (°C)	1.18	<0.001	1.06	<0.001
weather = adverse (3)	0.53	<0.001	0.62	<0.001
year = 2012	1.36	<0.001	1.63	<0.001

Workday status roughly halves casual demand (IRR 0.45) but has little marginal effect on registered demand once hour is controlled for; the registered commute signature lives in which hours are busy, not the daily total. Casual demand is also more weather-sensitive (IRR 0.53 vs. 0.62 in adverse weather) and grew more slowly year-over-year (IRR 1.36 vs. 1.63), consistent with membership-driven growth in registered ridership. Variable selection combined domain reasoning, nested-model AIC/F-tests, and a LassoCV robustness check that agreed with the chosen features.

Demand forecasting

Split (as prescribed): train = days 1–20 of each month (11,460 hours); test = day 21–month end (5,919 hours, 24 monthly windows). **Leakage prevention:** the longest test window is 264 hours, so any feature lag under 264 hours could reference a timestamp inside the window being predicted; all history-derived features use a 336-hour (2-week) lag or longer, verified with an executable check that no test-row feature reaches into its own test window. Two scenarios are reported: (A) strict, calendar and history features only (the realistic case); (B) oracle weather, which adds true future weather as an upper bound.

Model	Scenario	RMSLE	MAE	RMSE
Seasonal-mean baseline	A (strict)	0.651	79.4	118.9
Poisson GLM	A	0.580	54.0	90.2
Random Forest	A	0.536	47.3	80.5
LightGBM (Poisson)	A	0.531	47.5	81.5
LightGBM (Poisson)	B (oracle weather)	0.440	34.3	59.2

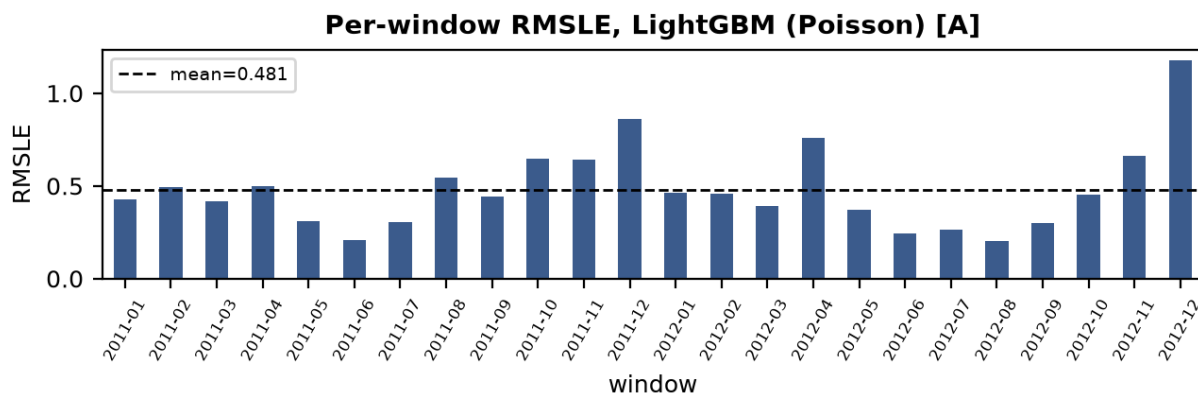


Figure 2. Per-window RMSLE across all 24 evaluation windows, best strict model.

- LightGBM with a Poisson objective is the best model in both scenarios, beating the seasonal-mean baseline by $\approx 18\%$ on RMSLE; knowing future weather improves RMSLE a further $\approx 17\%$, quantifying the cost of weather uncertainty in the realistic case.
- The worst windows (December of both years) coincide with the largest within-month demand-level shifts (winter onset, holidays), not the Hurricane Sandy/Irene storm windows, which a lag-based model handles fine since they are brief. A correlation of $r=0.75$ links per-window error to this level shift.
- A fully sequential rolling-origin backtest (training only on data prior to each window) gives RMSLE 0.544, close to the prescribed split's 0.531, supporting the headline result.

Part 2: Building Footprint Segmentation

Data & scope. Inria Aerial Image Labeling dataset: 5000×5000 px, 0.3 m RGB tiles with binary building masks. The official test split has no public masks, so the 180 labeled train tiles were partitioned independently. For time/compute reasons, a representative subset of 40 tiles (8 per city × 5 cities) was used, disclosed explicitly.

Data partitioning and leakage prevention

Each tile is cut into a non-overlapping 9×9 grid of 512×512 patches; no two patches can share a pixel, by construction, which rules out the overlapping-patch leakage mode by design. Two splits are used, both verified with executable assertions: a **primary split** (tile-disjoint, per-city-proportional: 6 train / 1 val / 1 test tile per city) for the headline result, and a **leave-one-city-out split** (Kitsap held out entirely, never seen in training or validation) as a direct, quantified probe of geographic leakage across cities.

Model and loss

U-Net with a ResNet34 encoder pretrained on ImageNet: skip connections preserve the fine edge detail a plain classifier would lose, and the pretrained encoder speeds convergence given the reduced data budget. The loss is **0.4×BCE + 0.4×Dice + 0.2×BoundaryDice**. Dice optimizes overlap and is robust to class imbalance (buildings cover 0.2%–46.4% of pixels depending on city); BCE gives a stable gradient; BoundaryDice is a Dice loss restricted to a narrow band around the ground-truth boundary, which targets edge precision directly in the loss rather than only measuring it afterward.

Quantitative results

Eval set	IoU	Dice	Precision	Recall	PixelAcc	BoundaryIoU
Primary (unseen tiles, known cities)	0.770	0.870	0.848	0.893	94.8%	0.564
Leave-one-city-out (Kitsap, never seen)	0.677	0.808	0.853	0.767	99.4%	0.518

Pixel accuracy alone would be misleading here (a trivial all-background prediction already scores 99.8% on the sparsest tile), which is why IoU, Dice, precision/recall, and boundary metrics are reported instead. Evaluating the same held-out city (Kitsap) once with a model that saw other tiles of that city and once with a model that never saw it at all shows an IoU drop of 0.139 (17% relative), the direct, quantified answer to the geographic-leakage question.

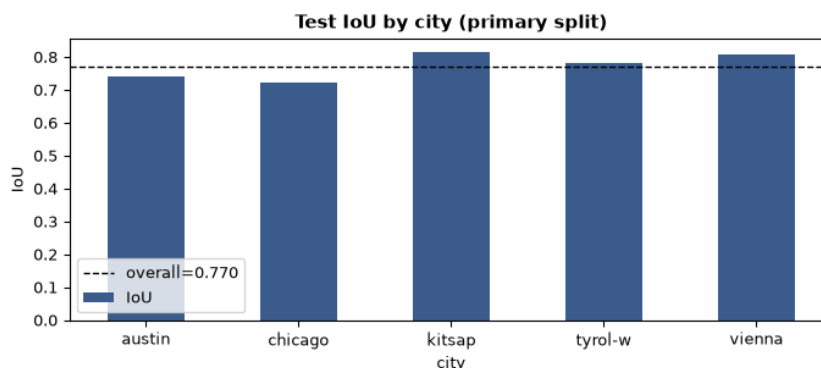


Figure 3. Test IoU by city (primary split).

Counter-intuitively, sparse/rural Kitsap has the highest in-distribution IoU of all five cities, not the lowest: isolated, high-contrast rooftops against forest/field are an easy case despite little training signal. Chicago and Vienna, the densest cities, are hardest in-distribution.

Qualitative visual assessment

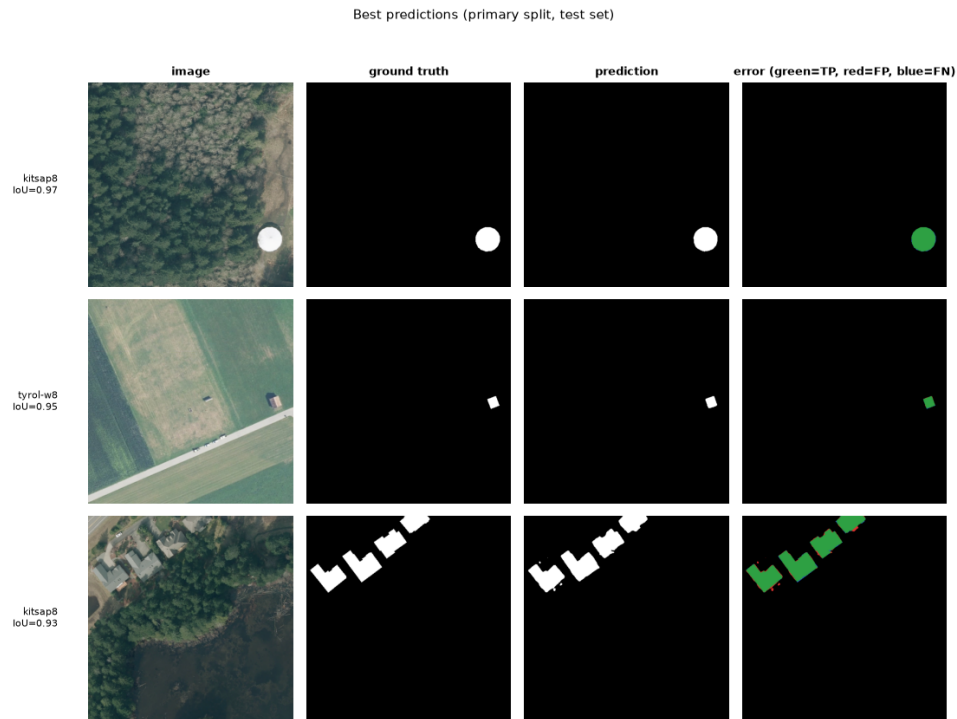


Figure 4. Best predictions: two Kitsap patches and one Tyrol-w patch, isolated rural buildings against a plain background.

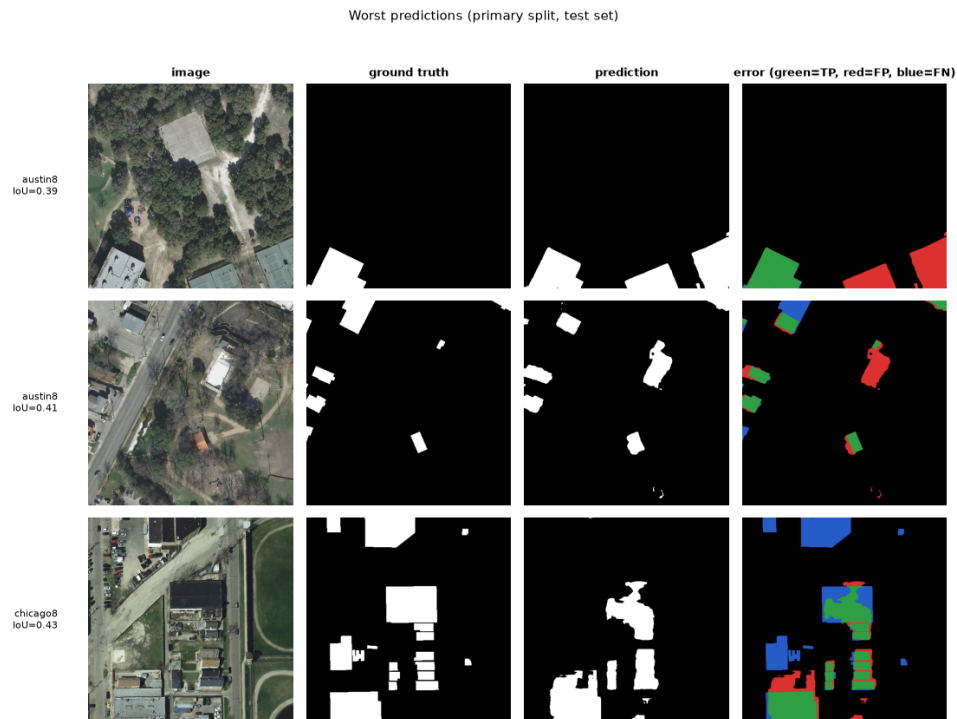


Figure 5. Worst predictions: Austin and Chicago. A large non-building structure mistaken for a rooftop, small structures only partially detected, and a dense complex with mixed correct and incorrect detections.

Adding the boundary term improved Boundary-IoU on every city relative to a BCE+Dice-only loss, confirming it had the intended effect rather than only adding complexity. Typical patches in dense cities are correct at the block level but scatter errors along the edges between adjacent buildings, a modest instance-adjacency effect expected of semantic (not instance) segmentation.

Conclusion

Both parts follow the same standard: leakage prevention verified by executable assertion rather than assumed, model and metric choices matched to the data's known statistical issues, and results reported with an honest breakdown across evaluation windows or cities rather than a single aggregate number. Part 1 delivers a leakage-safe forecasting model with RMSLE 0.531 on the prescribed split, confirmed by a rolling-origin backtest. Part 2 delivers a segmentation model with test IoU 0.770, and the leave-one-city-out experiment gives a concrete, quantified answer to the assessment's central geographic-leakage question: a 17% relative IoU drop when a city's visual style has never been seen.